

Big Cohen–Macaulay Algebras, Gabber’s Construction, and the Direct Summand Conjecture

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These are notes from my talk in the Winter 2026 almost mathematics seminar in Münster. This talk diverged from the rest of the seminar and didn’t use much of the previous material, so it should be fairly understandable to anyone with a reasonable commutative algebra background¹.

1. FROM DSC TO CM ALGEBRAS

The starting point here is Hochster’s Direct Summand Conjecture (DSC):

Theorem 1.1 (André). *Let A be a regular Noetherian ring and $f: A \rightarrow B$ a module-finite extension of rings. Then f admits a splitting in the category of A -modules.*

The seminar has been following Bhatt’s approach [Bha17], which is direct and based very much on almost mathematics and the theory of perfectoid spaces. Here we depart from Bhatt and explore an alternative path to proving the DSC, which was known to Hochster and ultimately also realized by André (in the same paper where he proved DSC, [And18]). The point of attack starts with a simplification to DSC. To achieve this, we introduce:

Definition 1.2 (Universally injective morphisms). *A morphism of rings $f: A \rightarrow B$ is called universally injective² if and only if $C \rightarrow B \otimes_A C$ is injective for every A -algebra (and hence A -module) C . Equivalently, f is universally injective if and only if the functor $- \otimes_A B: \text{Mod}_A \rightarrow \text{Mod}_B$ is faithful.*

It is apparent from the functorial definition that a morphism f is faithfully flat if and only if it is flat and universally injective. Moreover, if $f: A \rightarrow B$ and $g: B \rightarrow C$ are such that $g \circ f$ is universally injective, then f is universally injective. Combined with the following, we get our new approach to DSC.

Proposition 1.3. *Every finite and universally injective extension of Noetherian rings $f: A \hookrightarrow B$ is split.*

Proof. As splitting of f is equivalent to surjectivity of

$$\text{Hom}_A(B, A): g \mapsto g(1),$$

¹or at least a willingness to consult the Stacks project

²[Dos23] uses the term ‘pure’

we may assume A is local. Moreover, we can test this surjectivity after base change along the faithfully flat extension $A \rightarrow \hat{A}$; hence we assume A is complete as well. For E the injective hull of the residue field $k := A/\mathfrak{m}$, universal injectivity gives injectivity of the morphism $E \rightarrow E \otimes_A B$. As E is an injective A -module, the identity on E extends to an A -linear morphism $E \otimes_A B \rightarrow E$, which via $\otimes$ -Hom adjunction corresponds to an A -linear morphism $B \rightarrow \text{End}_A(E)$. However, $\text{End}_A(E) \simeq A$ by Matlis duality [Sta25, Tag 08Z1], so we have a splitting of f . 

In light of the above, to prove DSC it suffices to prove

Theorem 1.4. *For every finite ring extension $f: A \hookrightarrow B$ with A regular Noetherian, there exists a B -algebra C such that $A \rightarrow C$ is faithfully flat.*

Here is where Cohen–Macaulay algebras come in: they are our avenue to constructing such faithfully flat extensions.

Definition 1.5. *Let $(A, \mathfrak{m})$ be a Noetherian local ring and M an A -module.*

- (1) *A sequence $x_1, \dots, x_d$ in $\mathfrak{m}$ is called a system of parameters for A if and only if $d = \dim A$ and $A/(x_1, \dots, x_d)$ is Artinian (i.e. has dimension 0). Equivalently, $x_1, \dots, x_d$ is a system of parameters if and only if $d = \dim A$ and there exists an integer $t \geq 1$ such that $\mathfrak{m}^t \subset (x_1, \dots, x_d)$.*
- (2) *If M is an A -module, a sequence of elements $z_1, \dots, z_k$ in A is called M -regular or a regular sequence for M if and only if $M/(z_1, \dots, z_k)M \neq 0$ and multiplication by z_{i+1} is injective on $M/(z_1, \dots, z_i)M$ for each $0 \leq i < k$.*
- (3) *An A -module M is called (balanced big) Cohen–Macaulay (CM) if and only if every system of parameters in A is M -regular. A (balanced big) Cohen–Macaulay A -algebra is an A -algebra which is Cohen–Macaulay as an A -module. A itself is called a Cohen–Macaulay ring if it is a Cohen–Macaulay module over itself.*

Remark 1.6.

- (1) *In English literature, the term “Cohen–Macaulay” typically refers to a slightly different notion than we have defined here, and what we describe here are called “balanced³ big⁴ Cohen–Macaulay module/algebra.” However, in French literature, such as [And18] and [Dos23] (the main basis for this document), the term “Cohen–Macaulay” is typically used and ‘balanced big’ is omitted. As we are primarily following [Dos23], and in this setting Cohen–Macaulay algebras will tend to not be small, we will be lazy with terminology and simply say “Cohen–Macaulay.”*
- (2) *Whether a sequence $x_1, \dots, x_d$ is a system of parameters is clearly independent of ordering. On the other hand, whether such a sequence is M -regular for some M is a priori dependent*

³Balanced’ refers to the assumption that *every* system of parameters is M -regular (normally one only asks that *some* system of parameters is M -regular).

⁴Big’ refers to the fact that our modules are not necessarily finitely generated.

on the order of the elements. In the case that M is finitely generated⁵, being M -regular is actually independent of the order of the elements [Sta25, Tag 00LJ].

- (3) *A priori, our definitions of system of parameters and regular sequence allow for repeated elements in the sequence. However, such repetition cannot occur in a regular sequence due to the recursive non-zero-divisor criterion, and it cannot occur in a system of parameters due to Krull's Hauptidealsatz.*
- (4) *When M is finitely generated, there are a number of equivalent conditions to M being Cohen–Macaulay, most notably in terms of depth and vanishing of local cohomology; however in general these are not equivalent when M is not finite, and each leads to its own generalization of Cohen–Macaulayness for big modules.*

Example 1.7.

- (1) *Any Cohen–Macaulay ring is a Cohen–Macaulay algebra over itself. This class includes regular rings, Artinian rings, and reduced rings with dimension 1.*
- (2) *Cohen–Macaulay rings are stable under a number of operations: e.g. localisation, formation of polynomial/power series rings⁶, completion, certain quotients. Unfortunately, Cohen–Macaulay modules are generally not stable under such operations.⁷*
- (3) *If $A \hookrightarrow B$ is a local extension of local rings, then $\{x_1, \dots, x_d\} \subset A$ is a system of parameters for A if and only if it extends to a system of parameters for B . Similarly, if M is a B -module, then $x_1, \dots, x_d$ is M -regular if and only if it is M -regular when treating M as an A -module. Hence any CM B -module (or B -algebra) is also CM over A .*
- (4) *Any module over an Artinian ring is Cohen–Macaulay (there are no systems of parameters), and any faithfully flat module is Cohen–Macaulay (see theorem 1.10).*

Lemma 1.8. *Let $x_1, \dots, x_d$ and $y_1, \dots, y_d$ be two systems of parameters for a local Noetherian ring $(A, \mathfrak{m})$. There exist a system of parameters $v_1, \dots, v_d$ and a vector of nonnegative integers $(n_1, \dots, n_d) \in \mathbf{Z}_{\geq 0}^d$ such that*

$$(x_1^{n_1}, \dots, x_d^{n_d}) \subset (v_1, x_2^{n_2}, \dots, x_d^{n_d}) \subset \dots \subset (v_1, \dots, v_{d-1}, x_d^{n_d}) \subset (v_1, \dots, v_d) \\ \subset (v_1, \dots, v_{d-1}, y_d) \subset \dots \subset (v_1, y_2, \dots, y_d) \subset (y_1, \dots, y_d)$$

is a chain of $\mathfrak{m}$ -primary ideals in A .

Lemma 1.9. *Let M be an A -module and $z_1, \dots, z_k$ an M -regular sequence in A . Let $\mathfrak{z} := (z_1, \dots, z_k)$ and $\hat{M}$ the $\mathfrak{z}$ -adic completion of M . Then every permutation $y_1, \dots, y_k$ of the sequence $z_1, \dots, z_k$ is $\hat{M}$ -regular.*

Proof. Clearly $\hat{M}/(y_1, \dots, y_k)\hat{M} = M/\mathfrak{z}M \neq 0$ so it suffices to show that multiplication by y_i is injective on $\hat{M}/(y_1, \dots, y_{i-1})\hat{M}$ for each $1 \leq i \leq k$. Fix integers j, t such that $1 \leq j \leq k$

⁵As is often assumed when discussing Cohen–Macaulay modules.

⁶ $A[x]$ is not a local ring, but if we define a Cohen–Macaulay ring to be one which is Noetherian and all localizations at primes are Cohen–Macaulay local rings, then these all hold as stated.

⁷The obstructions are usually asking *every* system of parameters to be M -regular and requiring M regular sequences to satisfy $(z_1, \dots, z_k)M \neq M$.

and $t \geq 0$, and define

$$\begin{aligned} N_{j,t} &:= (y_1, \dots, y_{j-1}, y_j^t, \dots, y_k^t)M \\ N'_{j,t} &:= (y_1, \dots, y_{j-1}, y_j^{t+1}, y_{j+1}^t, \dots, y_k^t)M \\ M_{j,t} &:= (y_1, \dots, y_{j-1})M + \mathfrak{z}^t M. \end{aligned}$$

Clearly $N'_{j,t} \subset N_{j,t} \subset M_{j,t} \subset M$; for each t we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & M/N_{j,t} & \xrightarrow{y_j} & M/N'_{j,t} & \longrightarrow & M/N_{j+1,t} \longrightarrow 0 \\ & & \pi_{j,t} \downarrow & & \pi'_{j,t} \downarrow & & \downarrow \pi_{j+1,t} \\ & & M/M_{j,t} & \xrightarrow{y_j} & M/M_{j,t} & \longrightarrow & M/M_{j+1,t} \longrightarrow 0 \end{array}$$

where the vertical maps are the natural ones induced by the identity on M and the rows are exact. Fixing j , we get a projective system on the diagram indexed by t . As every morphism in each system is surjective, we get that the limit of the top row remains exact. Moreover, we have $M_{j,kt} \subset N_{j,t} \subset M_{j,t}$ for each t , hence the limit of the morphisms $\pi_{j,t}$ is an isomorphism. Similarly, the limit of the morphisms $\pi'_{j,t}$ is also an isomorphism, hence we have an exact sequence

$$0 \longrightarrow M_j \xrightarrow{y_j} M_j \longrightarrow M_{j+1} \longrightarrow 0$$

where $M_j := \varprojlim_t M/M_{j,t}$. But we can see that by construction $M_1 = \hat{M}$ and $M_k = \hat{M}/(y_1, \dots, y_{j-1})\hat{M}$. 

One of the remarkable features of balanced big Cohen–Macaulay modules is the following

Theorem 1.10 (Bartijn–Strooker, Hochster–Huneke). *Let $(A, \mathfrak{m})$ be a Noetherian local ring and M an A -module.*

- a) [BS83, Thm. 1.7] *If A has a system of parameters which is M -regular, then the $\mathfrak{m}$ -adic completion of M is CM over A .*
- b) [HH95, Thm. 2.1.d] *If A is regular, then M is CM if and only if it is faithfully flat.*

Proof. We follow the proofs given in [BS83] and [HH92, (6.7)].

- a) Let $x_1, \dots, x_d$ be an M -regular system of parameters and let $\mathfrak{x} := (x_1, \dots, x_d)$; by definition the $\mathfrak{x}$ -adic topology coincides with the $\mathfrak{m}$ -adic topology on M . By lemma 1.9, every $\hat{M}$ -regular sequence can be freely permuted. From lemma 1.8, it is enough to prove now that for a $\hat{M}$ -regular system of parameters $z_1, \dots, z_d$ and $w \in \mathfrak{m}$ such that $(z_1, \dots, z_d) \subset (z_1, \dots, z_{d-1}, w)$, the sequence $z_1, \dots, z_{d-1}, w$ is also $\hat{M}$ -regular. However, this is true for any A -module.
- b) Note that M is faithfully flat if and only if M is flat and $M/\mathfrak{m}M \neq 0$ [Sta25, Tag 00HP]. The second condition is equivalent to $M/(x_1, \dots, x_d)M$ for any system of parameters $x_1, \dots, x_d$. The first condition is equivalent to $\mathrm{Tor}_1^A(A/\mathfrak{p}, M) = 0$ for all

prime ideals $\mathfrak{p} \subset A$. Moreover, it is straightforward to inductively show that the following are equivalent:

- multiplication by x_{i+1} is injective on $M/(x_1, \dots, x_i)M$ for all $0 \leq i < k$;
- $\text{Tor}_1^A(A/(x_1, \dots, x_{i+1}), M) = 0$ for all $0 \leq i < k$;
- $\text{Tor}_j^A(A/(x_1, \dots, x_{i+1}), M) = 0$ for all $0 \leq i < k$ and all $j \geq 1$.

Clearly if M is flat this is satisfied, so any faithfully flat module is CM (without any assumption on A); it remains to show that any CM module over a regular local ring is flat. To this end, assume M is CM but not flat and choose $i \geq 1$ as large as possible so that there exists $\mathfrak{p} \in \text{Spec } A$ such that $\text{Tor}_i^A(A/\mathfrak{p}, M) \neq 0$ – such $\mathfrak{p}$ must exist as M is not flat and such maximal i must exist (and be bounded by $\dim A$) since A is regular. For any prime $\mathfrak{q} \in \text{Spec } A$ of depth ≥ 1 , we can construct a system of parameters $x_1, \dots, x_d$ for A such that $x_1 \in \mathfrak{q}$; as A is regular local it is a domain and hence our chosen prime $\mathfrak{p}$ has depth ≥ 1 . Hence we can construct an M -regular sequence in $\mathfrak{p}$; for any such sequence $x_1, \dots, x_k$, let $\mathfrak{x} := (x_1, \dots, x_k)$. We have a short exact sequence

$$0 \longrightarrow A/\mathfrak{p} \longrightarrow A/\mathfrak{x} \longrightarrow N \longrightarrow 0;$$

by taking a prime filtration of N [Sta25, Tag 00L0], we see that $\text{Tor}_{i+1}^A(N, M) = 0$, and by M -regularity of the sequence $(x_1, \dots, x_k)$, we also have $\text{Tor}_i^A(A/\mathfrak{x}, M) = 0$. Hence $\text{Tor}_i^A(A/\mathfrak{p}, M) = 0$ but this contradicts our choice of i and $\mathfrak{p}$. 🍌

In view of part b) of this theorem, we see that in theorem 1.4 we can change our focus from constructing faithfully flat extensions to constructing CM algebras. Moreover, part a) shows that by taking completions we need only construct algebras for which *some* system of parameters is regular (an “unbalanced” CM algebra). Hence we arrive at another conjecture of Hochster:

Theorem 1.11 (André). *For any local Noetherian ring $(A, \mathfrak{m})$, there exists a CM A -algebra.*

Proof of DSC assuming theorem 1.11. We prove theorem 1.4. Let A be a regular Noetherian ring and B a finite extension of A ; let $\mathfrak{m}$ be a maximal ideal of A and set $\hat{A}_{\mathfrak{m}} := \varprojlim_n A/\mathfrak{m}^n$, the $\mathfrak{m}$ -adic completion of A . Clearly $\hat{A}_{\mathfrak{m}} \rightarrow B \otimes_A \hat{A}_{\mathfrak{m}}$ remains finite; it may not remain injective, but we may choose a minimal prime $\mathfrak{p} \subset B \otimes_A \hat{A}_{\mathfrak{m}}$ such that $B' := (B \otimes_A \hat{A}_{\mathfrak{m}})/\mathfrak{p}$ has the same dimension as $B \otimes_A \hat{A}_{\mathfrak{m}}$. For such a prime, we see that $\mathfrak{p} \cap \hat{A}_{\mathfrak{m}} = 0$, as $\hat{A}_{\mathfrak{m}}$ is a domain [Sta25, Tag 00NP]. Hence $\hat{A}_{\mathfrak{m}} \rightarrow B'$ is a finite ring extension, and thus B' is local [Sta25, Tag 04GH]. We may hence choose a CM B' -algebra $C(\mathfrak{m})$, which is moreover CM over $\hat{A}_{\mathfrak{m}}$ and thus faithfully flat by theorem 1.10. As $\hat{A}_{\mathfrak{m}}$ is flat over A and A is Noetherian, it follows that

$$C := \prod_{\mathfrak{m} \in \text{MaxSpec } A} C(\mathfrak{m})$$

is a B -algebra which is flat over A . Moreover, we can see that by construction the map $\text{Spec } C \rightarrow \text{Spec } A$ is surjective on closed points, hence C is faithfully flat. 🍌

2. EXISTENCE OF CM ALGEBRAS

In our preceding discussion, we have reduced the proof of the direct summand conjecture to proving the existence of (balanced big) Cohen–Macaulay algebras over local Noetherian rings. To prove this, we can make some preliminary observations:

- As the $\mathfrak{m}$ -adic completion of a local ring $(A, \mathfrak{m})$ is a faithfully flat extension of A , we can immediately reduce theorem 1.11 to the case where A is complete.
- By choosing a minimal prime $\mathfrak{p} \in \text{Spec } A$ such that $\dim A/\mathfrak{p} = \dim A$, we may reduce to the case where A is a domain.

We now have three cases to consider:

Positive characteristic: $\text{char } A = p > 0$;

Equicharacteristic zero: $\text{char } A = \text{char } A/\mathfrak{m} = 0$;

Mixed characteristic: $\text{char } A = 0 < p = \text{char } A/\mathfrak{m}$.

The first case is rather explicitly resolved by Hochster–Huneke:

Theorem 2.1 ([HH92, Thm. 1.1]). *Let A be an excellent local domain of characteristic $p > 0$ with fraction field K , let $\bar{K}$ denote an algebraic closure of K , and let A^+ be the “absolute integral closure” of A i.e. the integral closure of A in $\bar{K}$. Then A^+ is a balanced big Cohen–Macaulay algebra over A .*

This theorem fails in equicharacteristic 0 when $\dim A \geq 3$; nevertheless, Hochster’s method of “algebra modifications” allows the existence of CM algebras in the equicharacteristic 0 case to be deduced from the characteristic p case [HH92, Thm. 8.1].

This leaves the third (and hardest) case: mixed characteristic. To handle this case, Dopinescu constructs an “almost perfectoid almost CM algebra,” then uses Gabber’s construction to convert this to an actual CM algebra.

Let B be a (necessarily non-Noetherian) A -algebra. Let $g \in B$ be “ p -powerful”⁸ i.e. nonzero and with a fixed system of compatible p^n -th power roots in B for all n . From now, we perform almost mathematics with respect to the ideal (g^{1/p^∞}) .

Definition 2.2 (Almost CM algebras). *A B -algebra C is called almost CM over A if there exists a system of parameters in A which is “almost C -regular.”*

A sequence $x_1, \dots, x_k$ in A is almost C -regular if and only if

- (1) $C/(x_1, \dots, x_k)C$ is not annihilated by (g^{1/p^∞}) i.e. it is not almost zero;
- (2) (g^{1/p^∞}) annihilates the C -module

$$\frac{((x_1, \dots, x_i)C : x_{i+1}C)}{(x_1, \dots, x_i)C}$$

*for each $0 \leq i < k$.*⁹

⁸[Bha17] calls these “perfect elements.”

⁹It was pointed out during the talk that this is in fact equivalent to multiplication by x_{i+1} being almost injective on $C/(x_1, \dots, x_i)C$ for each i .

Remark 2.3. A regular sequence need not be almost regular – clearly $C/(x_1, \dots, x_k)C$ can be almost zero but not 0. However, results of André imply that under some assumptions on C and g , regular sequences are indeed almost regular.

Gabber’s construction is a remarkably simple and elementary method of converting almost CM algebras to true CM algebras – much simpler than Hochster’s “partial modifications” that André originally used to accomplish this conversion. Let B be a ring and $g \in B$ p -powerful. Set

$$S := \left\{ (g^{a_n})_{n \geq 0} \in B^{\mathbf{N}} \mid a_n \in \mathbf{N} \left[\frac{1}{p} \right], \lim_{n \rightarrow \infty} a_n = 0 \right\}.$$

S is a multiplicative subset of the ring $B^{\mathbf{N}}$ not containing 0; hence for a B -module M , we can define the $B^{\mathbf{N}}$ -module $\mathcal{G}(M) := S^{-1}M^{\mathbf{N}}$. Note that this is also a B -module via the diagonal morphism $B \rightarrow B^{\mathbf{N}}$.

Lemma 2.4. If $f: M \rightarrow N$ is an almost injective map of B -modules, then the induced morphism $\mathcal{G}(f): \mathcal{G}(M) \rightarrow \mathcal{G}(N)$ is injective. Similarly, if $f: M \rightarrow N$ is an almost surjective map of B -modules, then $\mathcal{G}(f)$ is surjective.

Consequently, if $f: M \rightarrow N$ is an almost isomorphism of B -modules, then $\mathcal{G}(f)$ is an isomorphism.¹⁰

Proof.

Injectivity: $\mathcal{G}(f) \left(\frac{(m_n)_{n \geq 0}}{s} \right) = \frac{(f(m_n))_{n \geq 0}}{s}$, and this is 0 if and only if there exists $s' = (g^{a'_n})_{n \geq 0} \in S$ such that $s'(f(m_n)) = 0$. This holds if and only if $g^{a'_n} f(m_n) = 0$, and since f is almost injective, we find

$$g^{a'_n + 1/p^n} m_n = 0$$

for all n . But of course this implies that $\frac{(m_n)}{s} = 0 \in \mathcal{G}(M)$.

Surjectivity: Let $\frac{(k_n)}{s} \in \mathcal{G}(N)$; almost surjectivity of f shows that for all n there exists $m_n \in M$ for which $g^{1/p^n} k_n = f(m_n)$. But then

$$\mathcal{G}(f) \left(\frac{(m_n)}{s(g^{1/p^n})} \right) = \frac{(f(m_n))}{s(g^{1/p^n})} = \frac{k_n}{s}.$$



Proposition 2.5. Let C be a B -algebra and let $x_1, \dots, x_k$ be a sequence in B which is almost C -regular. Then $x_1, \dots, x_k$ becomes regular for $\mathcal{G}(C)$.

Proof. We follow the argument in [Dos23]; it was pointed out during the talk that since condition (2) in the definition of an almost C -regular sequence is equivalent to almost injectivity of multiplication by x_{i+1} , lemma 2.4 implies the corresponding (non-almost) condition for $\mathcal{G}(C)$.

¹⁰According to Kostas, these can all be strengthened to if and only ifs, with the caveat that we need to talk about a morphism f inducing $\mathcal{G}(f)$. In general, two B -modules can satisfy $\mathcal{G}(M) \simeq \mathcal{G}(N)$ without M and N being almost isomorphic.

We first show that $\mathcal{G}(C)/(x_1, \dots, x_k)\mathcal{G}(C) \neq 0$. Supposing otherwise, there exist $s \in S$ and $y_1, \dots, y_k \in C^N$ such that

$$1 = \sum_{i=1}^k x_i \frac{y_i}{s}.$$

This holds if and only if there exists $s' \in S$ such that

$$s' \left(s - \sum_{i=1}^k x_i y_i \right) = 0.$$

Writing $s = (g^{a_n})_{n \geq 0}$, $s' = (g^{a'_n})_{n \geq 0}$ and projecting to the n th component, we deduce that $g^{a_n + a'_n} \in (x_1, \dots, x_k)C$ for each n . As $a_n + a'_n \rightarrow 0$, we see that (g^{1/p^∞}) annihilates $C/(x_1, \dots, x_k)C$, which is a contradiction.

Now we wish to verify that if $a \in \mathcal{G}(C)$ and $ax_{i+1} \in (x_1, \dots, x_i)\mathcal{G}(C)$, then $a \in (x_1, \dots, x_i)\mathcal{G}(C)$. For this, we write $a = \frac{b}{s}$ with $b \in C^N$ and $s \in S$ such that there exist $z_1, \dots, z_i$ so that

$$ax_{i+1} = \sum_{j=1}^i x_j \frac{z_j}{s}.$$

Then there exists $s' = (g^{a'_n})_{n \geq 0} \in S$ such that

$$s' \left(bx_{i+1} - \sum_{j=1}^i x_j z_j \right) = 0;$$

projecting to the n th component shows $g^{a'_n} b_n x_{i+1} \in (x_1, \dots, x_i)C$ for each n . As (g^{1/p^∞}) annihilates

$$(1) \quad \frac{((x_1, \dots, x_i)C : x_{i+1}C)}{(x_1, \dots, x_i)C},$$

we see that $g^{a'_n + 1/p^n} b_n \in (x_1, \dots, x_i)C$ for each n . Hence we may write

$$g^{a'_n + 1/p^n} b_n = \sum_{j=1}^i x_j u_{n,j}$$

for some $u_{n,j} \in C$. Setting $u_j := (u_{n,j})_{n \geq 0} \in C^N$ and $s'' := (g^{1/p^n})_{n \geq 0}$, we have

$$s' s'' b = \sum_{j=1}^i x_j u_j,$$

and hence

$$a = \frac{b}{s} = \sum_{j=1}^i x_j \frac{u_j}{ss's''} \in (x_1, \dots, x_i)\mathcal{G}(C).$$



REFERENCES

- [And18] Yves André. La conjecture du facteur direct. *Publ. Math. Inst. Hautes Études Sci.*, 127:71–93, 2018.
- [Bha17] Bhargav Bhatt. Lecture notes for a class on perfectoid spaces. <https://www.math.ias.edu/~bhatt/teaching/mat679w17/lectures.pdf>, April 2017.
- [BS83] Jaap Bartijn and Jan R. Strooker'. Modifications monomiales. In Marie-Paule Malliavin, editor, *Séminaire d'Algèbre Paul Dubreil et Marie-Paule Malliavin*, pages 192–217, Berlin, Heidelberg, 1983. Springer Berlin Heidelberg.
- [Dos23] Gabriel Dospinescu. La conjecture du facteur direct [d'après André et Bhatt]. Number 446, pages Exp. No. 1201, 141–197. 2023. Séminaire Bourbaki. Vol. 2022/2023. Exposés 1197–1210.
- [HH92] Melvin Hochster and Craig Huneke. Infinite integral extensions and big cohen–macaulay algebras. *Annals of Mathematics*, 135(1):53–89, 1992.
- [HH95] M. Hochster and C. Huneke. Applications of the existence of big cohen–macaulay algebras. *Advances in Mathematics*, 113(1):45–117, 1995.
- [Sta25] The Stacks project authors. The stacks project. <https://stacks.math.columbia.edu>, 2025.